The Rise of Generative AI in Online Discourse: A Reddit-Based Social Media Analysis

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Abstract

Generative artificial intelligence (AI) has rapidly transitioned from a research innovation to an everyday digital infrastructure used for writing, programming, image generation, and decision support. As tools such as ChatGPT, GPT-4, Midjourney, and Stable Diffusion become widely adopted, public perception and user experience increasingly shape how these systems are trusted, evaluated, and integrated into daily workflows. This study examines how generative AI is discussed across online communities by analyzing posts and comments from multiple AI-focused subreddits on Reddit. Using a combination of lexicon-based sentiment analysis (VADER) and transformer-based topic modeling (BERTopic), the analysis captures both emotional tone and thematic structure at the community level. The results show a clear divergence between optimistic post framing and more critical, problem-oriented comment discourse, as well as substantial variation across subreddits. Topic modeling reveals that discussions are dominated by practical concerns such as troubleshooting, usability, and workflow integration, while ethical and societal topics appear as smaller but persistent themes. Temporal analysis further indicates a sharp increase in discussion volume during late 2025, coinciding with broader adoption of generative AI tools. Overall, the findings demonstrate that generative AI is increasingly treated not as a speculative technology but as a practical, infrastructure-like system, with community norms playing a central role in shaping sentiment and discourse.

1. Introduction

Generative artificial intelligence (AI) has rapidly evolved from a research concept into a widely deployed set of tools that influence everyday digital practices. One of the most significant technological advancements of the past decade, generative AI is now widely used for writing, programming, image creation, education, and decision assistance through systems such as ChatGPT, GPT-4, Midjourney, and Stable Diffusion. As these systems become increasingly embedded in daily workflows, their adoption, trust, and regulation are shaped not only by technical performance but also by user experiences and public perception, a pattern consistently observed in prior studies of artificial intelligence and emerging technologies (Fast & Horvitz, 2017; Liu & Wang, 2020). To anticipate broader societal implications, identify points of contention, and inform responsible AI development, it is therefore critical to understand how people discuss and respond to generative AI in online environments.

Online social platforms provide a valuable lens for examining these dynamics because they capture spontaneous, user-driven discourse at scale. Research in political communication and computational social science has shown that online platforms do not merely reflect public opinion but actively shape how information is interpreted, amplified, and contested through social interaction (Iyengar & Hahn, 2009; Flaxman, Goel, & Rao, 2016). Reddit, in particular, hosts large, topic-focused communities where users openly share opinions, experiences, and technical knowledge through long-form posts and threaded discussions. Unlike short-form platforms such as Twitter, Reddit interactions frequently involve detailed explanations, collaborative troubleshooting, and extended debate, making the platform especially well suited for studying how users collectively engage with complex technologies like generative AI.

Previous studies have examined public reactions to automation, artificial intelligence, and other emerging technologies, often focusing on aggregate sentiment trends or high-level thematic shifts over time. More recent work on generative AI has identified a transition from early technical curiosity toward broader societal concerns related to ethics, trust, and impact (Miyazaki, Sato, & Nagata, 2024). However, much of this literature relies on short-form platforms, limited temporal windows, or analyses that treat public discourse as relatively homogeneous. As a result, we still lack a clear understanding of how different online communities frame generative AI differently, how sentiment diverges between original posts and interactive comment threads, and which concrete topics dominate everyday user discussions during periods of rapid adoption.

To address these gaps, this study conducts a community-level analysis of generative AI discourse on Reddit using a newly collected dataset of posts and comments from multiple AI-focused subreddits. By combining lexicon-based sentiment analysis with transformer-based topic modeling, this work moves beyond aggregate trends to examine how emotional tone and thematic focus vary across communities and interaction contexts, building on prior findings that emphasize the role of interaction dynamics in shaping online sentiment (Cheng et al., 2017). In doing so, the study provides empirical evidence that generative AI is increasingly discussed not only as a transformative technology, but also as a practical, infrastructure-like tool—one that users actively rely on, critique, and collectively refine through ongoing online engagement.

2. Related work

Miyazaki, Sato, and Nagata (2024) conducted a large-scale analysis of more than three million Twitter posts to examine how public perception of generative AI evolved between 2019 and 2023. Their study found a clear shift in discourse following the release of ChatGPT, with conversations moving from technical curiosity toward ethical, social, and political concerns. This work is highly relevant to the present study because it establishes that generative AI discourse changes rapidly in response to technological milestones. However, their analysis relies on short-form Twitter data and focuses primarily on aggregate sentiment trends. In contrast, the current project examines long-form Reddit discussions and demonstrates that, at the community level, generative AI discourse is dominated by practical usage issues such as troubleshooting, system failures, and workflow integration—an aspect that is largely absent from Twitter-based analyses.

Fast and Horvitz (2017) analyzed long-term public perceptions of artificial intelligence using online text data spanning multiple decades. They showed that public attention toward AI fluctuates between optimism and fear, often in response to major technological breakthroughs. Their findings highlight how societal narratives around AI are shaped by broader cultural and technological contexts. This study is directly relevant because it provides a historical foundation for understanding emotional reactions to AI. While Fast and Horvitz emphasize macro-level trends over time, the present study complements their work by focusing on micro-level, community-specific discourse. Rather than fear or hype alone, Reddit discussions reveal a more pragmatic orientation, suggesting that generative AI has transitioned from a speculative technology to an everyday digital tool.

Liu and Wang (2020) investigated how emerging technologies diffuse through social media by analyzing patterns of public acceptance and discussion across online communities. They found that social media plays a critical role in shaping the legitimacy of new technologies, as user discussions influence whether innovations are perceived as beneficial, disruptive, or threatening. This work is closely connected to the current project's focus on subreddit-level differences. Consistent with Liu and Wang's findings, the present study shows that different Reddit communities form distinct "AI microcultures," each emphasizing different concerns—ranging from creativity and experimentation to skepticism and technical reliability. This reinforces the idea that public acceptance of generative AI is not uniform but socially constructed within specific online contexts.

Cheng et al. (2017) examined how user behavior and sentiment evolve within online discussion communities, showing that negative sentiment and conflict often emerge through interaction rather than initial posts. Their findings are highly relevant to the present study's observation that Reddit comments tend to be more negative than posts. This work supports the interpretation that sentiment polarization in generative AI discussions arises from community interaction dynamics rather than individual attitudes alone. Importantly, their results highlight that sentiment trajectories within online discussions are not static but evolve as conversations grow in depth and participation. This observation aligns with the present study's methodological choice to analyze both posts and comments, enabling a clearer distinction between initial framing and interaction-driven sentiment dynamics in generative AI discourse. By applying this insight to AI-related subreddits, the current study extends prior findings to the context of emerging technologies and demonstrates how collective discussion shapes public perception of generative AI tools.

Zannettou et al. (2018) investigated how online communities shape and transform discourse around emerging topics, demonstrating that community norms strongly influence how information is framed and propagated. Although their study focuses on memes rather than AI, it provides a valuable framework for understanding how distinct online communities develop unique interpretive lenses. Their work emphasizes that community-specific cultures can amplify certain narratives while suppressing others, even when discussing the same underlying topic. This insight supports the present study's finding that generative AI is framed differently across subreddits, underscoring the importance of community context when interpreting social media discussions around emerging technologies. This perspective directly informs the present study's community-level analysis of generative AI discussions on Reddit, where subreddits exhibit distinct thematic priorities and emotional tones despite discussing the same underlying technology.

3. Data Collection

The dataset used in this study was entirely self-collected from Reddit, a large-scale social media platform that hosts topic-focused discussion communities known as subreddits. Reddit was chosen because it provides rich, publicly accessible, user-generated content in the form of posts and threaded comments, making it particularly well-suited for studying collective discourse, opinion formation, and interaction dynamics around emerging technologies. The platform's community-based structure enables comparisons across distinct social contexts, which is central to understanding how generative artificial intelligence (AI) is framed and discussed by different online groups.

Data collection was conducted using direct web requests to Reddit's platform-provided JSON endpoints, rather than relying on any pre-existing datasets or third-party data dumps. Specifically, the /new.json endpoint was queried for multiple AI-related subreddits, including *r/ChatGPT*, *r/Midjourney*, *r/StableDiffusion*, *r/MachineLearning*, and *r/ArtificialIntelligence*. This endpoint returns the most recent submissions in each subreddit, allowing the collection of timely and organically generated discussions. Requests were issued with a custom user-agent header and controlled delays between requests to comply with Reddit's access policies and avoid rate-limiting.

To ensure relevance to generative AI, posts were filtered using a predefined set of keywords related to large language models and image-generation systems, including terms such as *ChatGPT*, *GPT-4*, *Midjourney*, *Stable Diffusion*, *DALL·E*, *large language model*, and *generative AI*. Only posts whose titles or body text contained at least one of these keywords were retained. The collection period focused on posts published between January 2025 and the time of data collection, capturing contemporary discourse during a period of rapid public adoption and debate surrounding generative AI tools. In total, this process yielded over 5,000 unique posts after deduplication.

In addition to posts, the study collected user comments associated with a subset of highly engaged posts to capture interaction-driven discourse. For each selected post, its corresponding discussion thread was retrieved by appending .json to the post's URL, which returns both the submission metadata and the complete hierarchical comment structure. A recursive parsing procedure was used to flatten nested comment trees into individual comment records while preserving metadata such as parent-child relationships, author information, timestamps, and community identifiers. To maintain computational feasibility and comply with rate limits, an upper bound was placed on the number of comments retrieved per post, and automatic backoff strategies were implemented in response to HTTP 429 (Too Many Requests) errors. This process resulted in a corpus of over 100,000 unique comments.

The final dataset consists of two primary components: a post-level dataset containing textual content, timestamps, subreddit identifiers, engagement metrics, and metadata; and a comment-level dataset capturing user responses, interaction depth, and temporal dynamics. Together, these datasets form a large-scale, original social media corpus that reflects how generative AI is discussed, evaluated, and contested within online communities. All data collection scripts, parameters, and preprocessing steps are fully documented and reproducible, ensuring transparency and enabling future extensions of the dataset for additional analyses.

During the course of this project, Reddit's official API access policies underwent changes that introduced stricter approval requirements and rate limitations for newly registered applications. As a result, obtaining immediate authenticated API credentials was not feasible within the project timeline. To ensure compliance while maintaining data integrity, the data collection strategy was adapted to rely on Reddit's publicly available JSON endpoints, which are explicitly designed to support programmatic access to public content. This approach does not circumvent platform safeguards and has been widely used in prior computational social science research for collecting publicly accessible Reddit discussions. By leveraging these endpoints with conservative request rates and custom user-agent headers, the study ensured ethical data collection while preserving reproducibility and transparency.

4. Methods

This study employs a multi-stage social media mining pipeline to analyze public discourse surrounding generative artificial intelligence on Reddit. The analytical approach combines large-scale text preprocessing, sentiment analysis, and topic modeling to capture both the emotional tone and thematic structure of online discussions. All analyses were conducted using Python, with key libraries including pandas, nltk, scikit-learn, sentence-transformers, and BERTopic.

Following data collection, the raw dataset was first cleaned and standardized. For posts, the title and body text were concatenated to form a unified textual representation, while for comments, only the comment body was retained. Non-informative content such as deleted or removed posts and comments was excluded. Timestamps were converted from Unix format to timezone-aware datetime objects to enable temporal analysis. Additional features, including text length and keyword presence flags, were computed to support exploratory analysis and filtering. To focus specifically on generative AI discourse, a curated list of domain-relevant keywords (e.g., "ChatGPT," "GPT-4," "Midjourney," "Stable Diffusion," and "large language model") was used to identify posts and comments explicitly referencing generative AI tools.

Sentiment analysis was conducted using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model, which is well-suited for short, informal social media text. For each post and comment, VADER's compound sentiment score was computed and used as the primary sentiment indicator, capturing polarity on a continuous scale from negative to positive.

Sentiment distributions were analyzed across subreddits, over time, and between posts and comments to assess how emotional tone varies by community and interaction type. Visualization techniques such as boxplots, histograms, and temporal trend plots were used to summarize these patterns.

To uncover the latent thematic structure of generative AI discussions, topic modeling was performed using BERTopic. This method combines transformer-based sentence embeddings with clustering and class-based TF-IDF to generate semantically coherent topics. Sentence embeddings were generated using the pre-trained “all-MiniLM-L6-v2” model, and topic representations were derived using a CountVectorizer with frequency-based filtering to remove overly rare and overly common terms. Topic modeling was applied to a subset of posts that explicitly mentioned generative AI keywords to ensure thematic relevance. Each post was assigned a topic label, allowing for analysis of topic prevalence across subreddits and over time.

5. Findings

Figure 1 shows how comment activity related to generative AI is distributed across different Reddit communities, revealing a clear imbalance in where discussion is most intense. General-interest subreddits such as *r/technology* and *r/Futurology* dominate in terms of comment volume, indicating that generative AI is widely discussed in mainstream technology spaces rather than being confined to niche or expert-only communities. In contrast, more specialized subreddits like *r/ChatGPT*, *r/OpenAI*, and *r/ArtificialIntelligence* generate fewer comments, suggesting that while these spaces support focused, technical, and problem-solving discussions, they attract smaller and more targeted audiences. Overall, the figure highlights that public engagement with generative AI is driven largely by broad, socially oriented communities, whereas specialized forums play a complementary role by hosting deeper but less populous discussions, underscoring the need to analyze discourse across multiple community types.

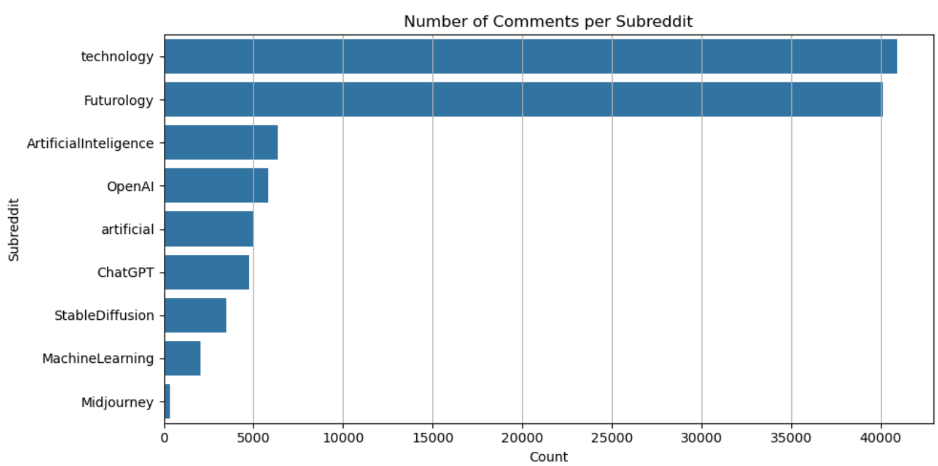

Figure 1. Distribution of Comment Volume Across AI-Related Subreddits

Figure 2 compares how often generative AI is explicitly referenced in original posts versus comment replies across the collected Reddit dataset. The results show that a substantially larger proportion of posts directly mention generative AI keywords compared to comments, indicating that AI is typically introduced as the central topic at the post level. In contrast, comments are far less likely to explicitly name AI tools, suggesting that once a discussion is initiated, participants tend to focus on contextual responses such as troubleshooting, opinions, or follow-up questions without repeatedly referencing the technology itself. This pattern highlights a structural difference in Reddit discourse, where posts serve to frame AI-related conversations, while comments reflect deeper, more implicit engagement with the topic.

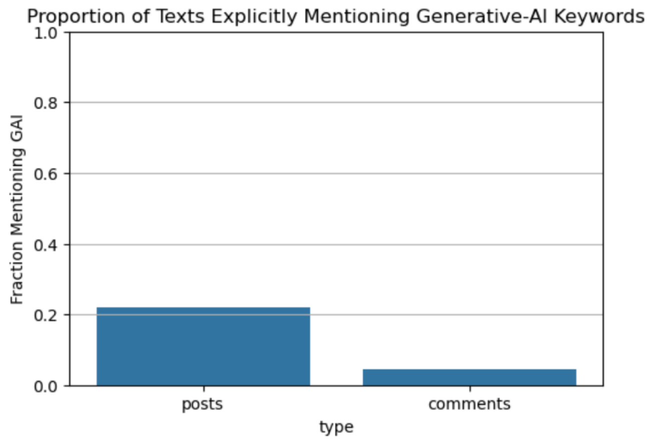

Figure 2. Proportion of Posts and Comments Explicitly Mentioning Gen-AI Keywords

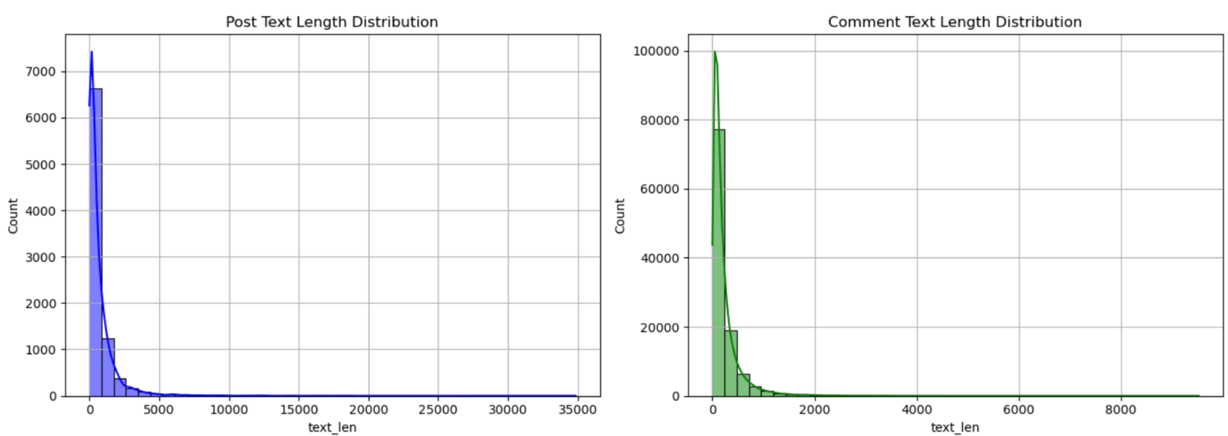

Figure 3. Text Length Distributions for Posts and Comments

Figure 3 shows the distribution of text lengths for Reddit posts and comments in the dataset, highlighting structural differences in how users initiate versus participate in generative AI discussions. The post-length distribution is right-skewed, with most posts being relatively short but a long tail of very lengthy posts, indicating that users often introduce AI-related topics concisely while occasionally providing detailed explanations, use cases, or problem descriptions. In contrast, comment lengths are much more tightly concentrated around shorter texts, reflecting the conversational nature of replies where users tend to provide brief reactions, clarifications, or troubleshooting suggestions rather than extended narratives. Together, these distributions suggest that posts primarily serve as framing mechanisms for AI discussions, while comments function as iterative, focused contributions that collectively drive the depth of conversation.

Figure 4 compares the distribution of VADER compound sentiment scores for Reddit posts and comments related to generative AI, revealing a clear divergence in emotional tone between content initiation and community response. Posts show a stronger concentration toward positive sentiment, with a noticeable peak at high positive values, suggesting that users often frame generative AI in optimistic terms when sharing updates, success stories, or new capabilities. In contrast, comments are more tightly centered around neutral and mildly negative sentiment, with a broader spread into negative values, indicating that discussion threads frequently evolve into troubleshooting, critique, and debate. This pattern suggests that while generative AI is introduced in an upbeat or hopeful manner, collective interaction within comment threads surfaces frustrations, limitations, and skepticism, highlighting how sentiment polarization emerges through social interaction rather than initial framing alone.

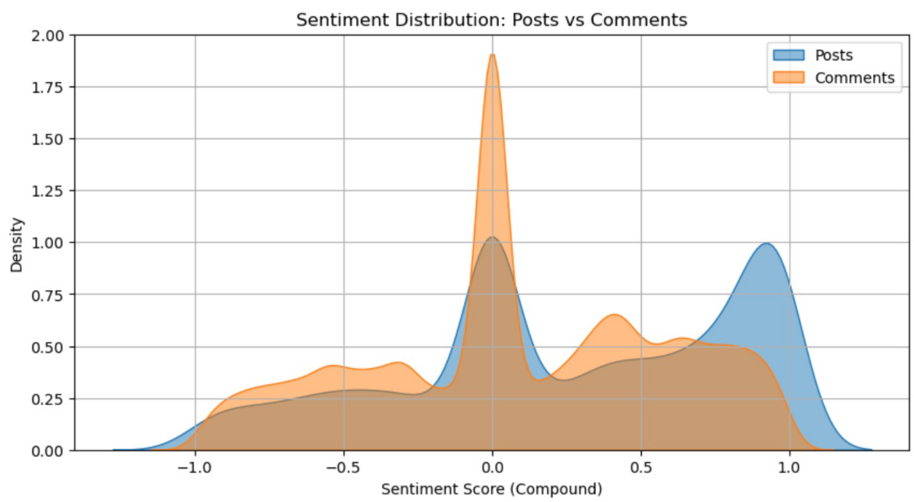

Figure 4. Sentiment Distribution of Posts vs. Comments

Figure 5 shows the distribution of VADER sentiment scores for generative-AI–related posts across different Reddit communities, highlighting clear community-level differences in how AI is framed. Technical and practitioner-oriented subreddits such as *MachineLearning*, *ArtificialIntelligence*, and *StableDiffusion* exhibit higher median sentiment values, indicating generally optimistic or constructive discussions focused on usage, performance, and experimentation. In contrast, broader or more general subreddits such as *technology* and *ChatGPT* show lower medians and tighter distributions around neutral to mildly negative sentiment, reflecting greater skepticism, critical evaluation, and discussion of limitations or concerns. Importantly, none of the subreddits span the full sentiment range from highly negative to highly positive; instead, each community maintains a relatively bounded emotional profile. This suggests that sentiment toward generative AI is shaped less by extreme polarization and more by stable community norms, reinforcing the idea that different online communities develop distinct “AI microcultures” in how they discuss and evaluate emerging technologies.

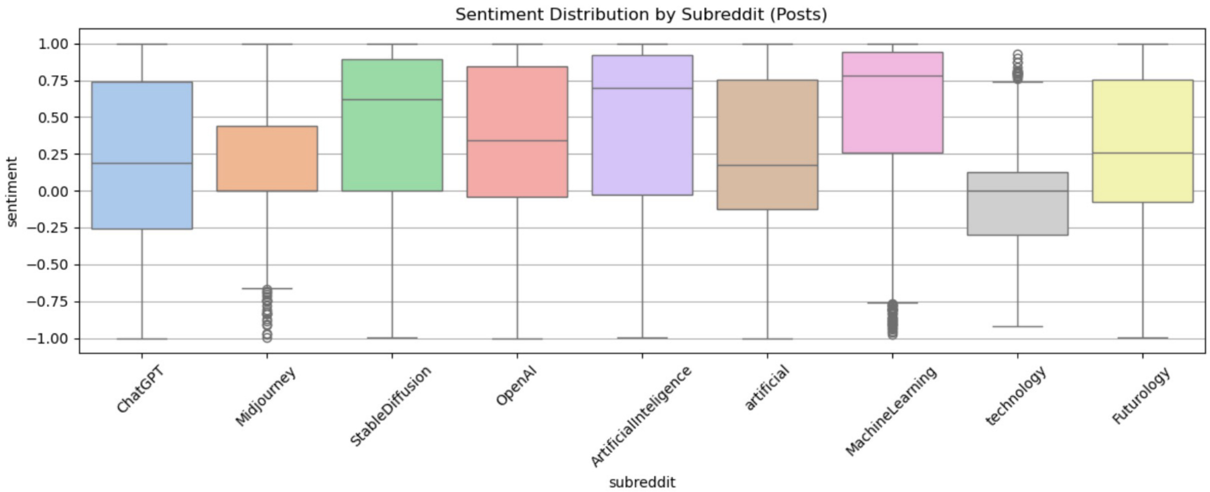

Figure 5. Sentiment Distribution by Subreddit (Posts)

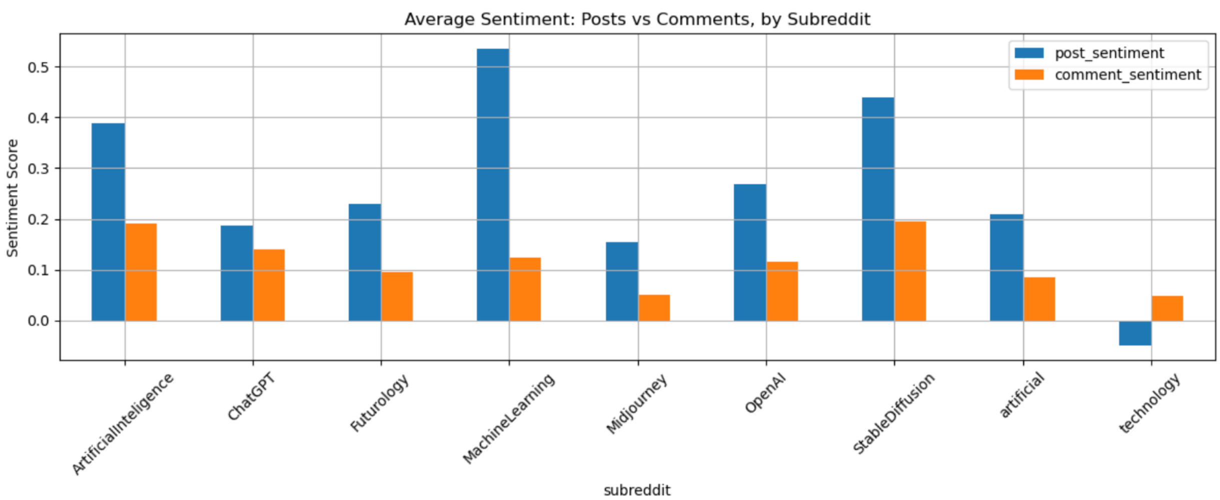

Figure 6. Average Sentiment Distribution of Posts vs. Comments, by Subreddit

Figure 6 compares the average VADER sentiment scores of original posts and their corresponding comments across AI-related subreddits, revealing a consistent and important pattern. Across all communities, posts are systematically more positive than comments, indicating that users tend to introduce generative AI topics with optimism, curiosity, or success-oriented framing, while comment threads shift toward more critical, corrective, or problem-solving discourse. The gap is especially pronounced in technically focused communities such as *MachineLearning* and *StableDiffusion*, where posts are highly positive but comments are substantially less so, reflecting rigorous peer evaluation and troubleshooting norms. In contrast, broader subreddits like *technology* and *artificial* show relatively low sentiment in both posts and comments, suggesting a more skeptical baseline attitude toward generative AI. Overall, this divergence highlights that sentiment polarization around generative AI is not driven by initial opinions alone, but emerges through interaction, discussion, and collective sense-making within online communities.

Figure 7 presents the top generative-AI discussion topics extracted using BERTopic, ranked by the number of posts assigned to each topic. The results show that the most prevalent topics are overwhelmingly practical and use-oriented rather than abstract or speculative. The largest clusters focus on issues such as human–AI interaction in real-world applications, fixing ChatGPT or image-generation errors, style and reference image creation, GitHub/model training and reasoning challenges, and OpenAI account or email access problems. These high-volume topics indicate that users primarily engage with generative AI as a tool embedded in everyday workflows, where troubleshooting, optimization, and usability concerns dominate discussion.

At the same time, the topic distribution in Figure 7 reveals substantial thematic diversity beyond the largest clusters. Smaller but persistent topics related to safety, mental health, ethical concerns, environmental impact, and comparative evaluations between models (e.g., ChatGPT vs. Gemini) suggest that broader societal implications are present but secondary to operational issues. The long tail of mid-sized and small topics highlights that generative AI discourse does not converge around a single narrative; instead, it spans many niche, task-specific, and context-dependent conversations. Together, these findings support the conclusion that public engagement with generative AI on Reddit is driven less by futuristic speculation and more by hands-on problem solving, while still leaving room for emerging ethical and social debates.

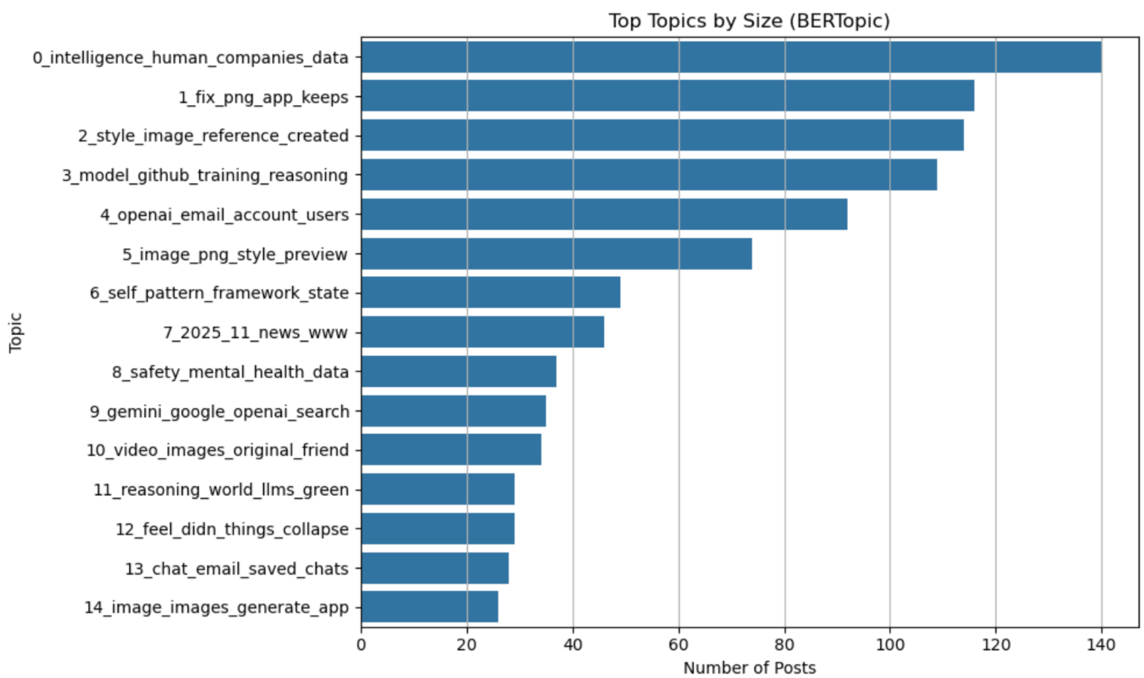

Figure 7. Dominant Discussion Topics in Generative AI Discourse Identified by BERTopic

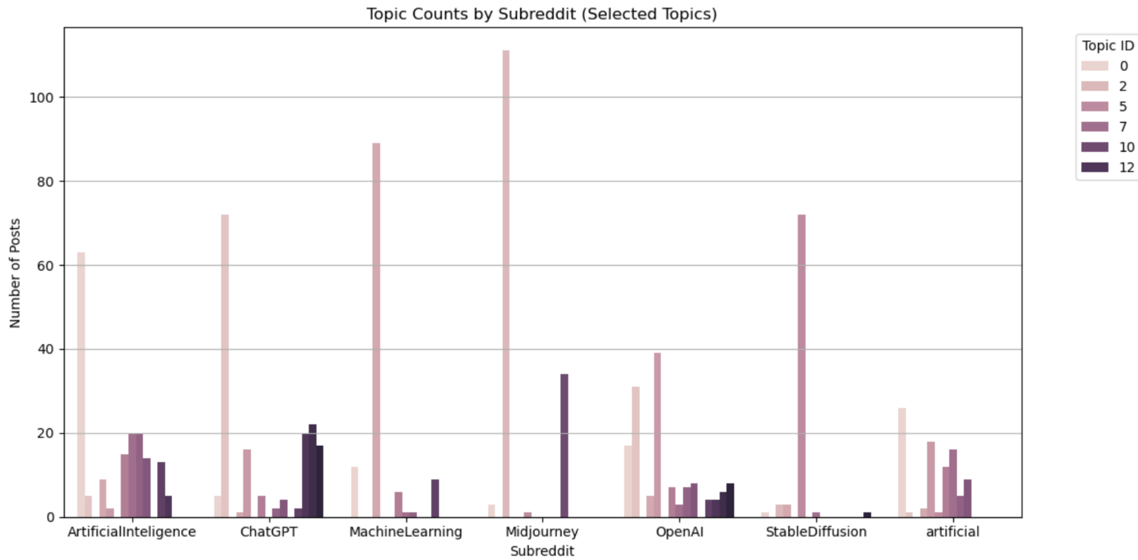

Figure 8. Community-Level Distribution of Key Generative AI Topics Across Subreddits

Figure 8 illustrates clear community-level differences in how key generative AI topics are discussed across Reddit subreddits. While similar topics appear in multiple communities, their relative prominence varies substantially: r/MachineLearning and r/ArtificialIntelligence emphasize model reasoning, training, and technical performance, reflecting expert-driven discourse, whereas r/Midjourney and r/StableDiffusion are heavily centered on image generation, style, and creative workflows. Subreddits such as r/ChatGPT and r/OpenAI exhibit a mixed pattern, combining practical troubleshooting with discussions about access, usability, and platform constraints. Overall, the results show that generative AI discourse on Reddit is not uniform but shaped by distinct community norms and purposes, supporting the idea that different subreddits function as specialized “AI microcultures” that frame the same technology in fundamentally different ways.

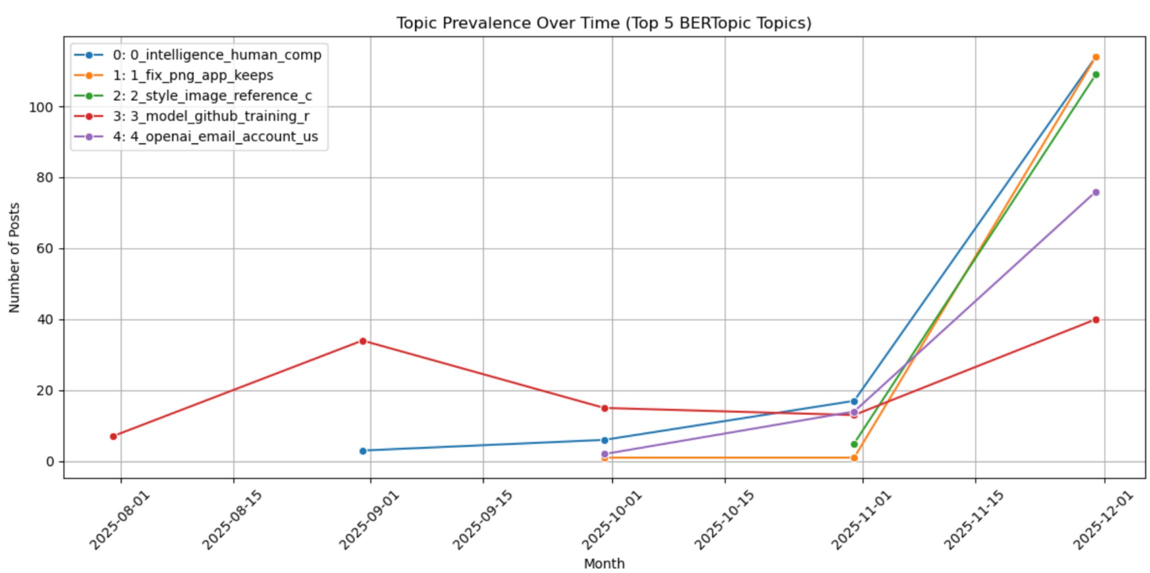

Figure 9. Topic Prevalence over Time

Figure 9 shows a clear temporal shift in the nature and intensity of generative AI discussions over the second half of 2025. Across all top five BERTopic topics, discussion volume remains relatively low and fragmented from August through October, indicating early-stage or sporadic engagement. However, beginning in November 2025, there is a sharp and simultaneous surge across multiple topics—particularly practical themes such as troubleshooting ChatGPT behavior, fixing image-generation issues, handling style/reference images, and managing OpenAI account or access problems. This coordinated rise suggests that generative AI tools transitioned from experimental or niche usage to widespread, routine adoption during this period, triggering increased hands-on problem-solving and user support discussions. The comparatively smaller but steady growth of model training and GitHub-related topics further indicates that while advanced technical conversations persist, the dominant driver of discourse is everyday operational use, reinforcing the finding that generative AI has become embedded in daily digital workflows rather than remaining a purely exploratory technology.

6. Conclusion

This study examined how generative artificial intelligence is discussed across multiple Reddit communities by combining large-scale social media data collection with sentiment analysis and transformer-based topic modeling. The findings show that generative AI discourse on Reddit is highly active, community-driven, and shaped by practical engagement rather than abstract speculation. Discussion volume increased sharply during the latter part of 2025, reflecting the rapid normalization of tools such as ChatGPT and Stable Diffusion in everyday workflows. Across subreddits, posts tend to frame generative AI in optimistic or exploratory terms, while comment threads shift toward troubleshooting, critique, and debate, revealing a consistent divergence between initial framing and interactive discourse.

At the thematic level, BERTopic reveals that the dominant topics are overwhelmingly practical, focusing on fixing errors, improving image generation, resolving account and login issues, and evaluating model performance. Philosophical or futuristic discussions appear far less frequently, suggesting that generative AI is increasingly perceived as an operational technology rather than a speculative one. The presence of a large outlier topic further highlights the diversity and fragmentation of AI-related discussions, indicating that many user concerns do not neatly fit into predefined categories. In addition, smaller but persistent topics related to safety, ethics, and mental health demonstrate that societal concerns remain embedded within technical conversations, even if they do not dominate discussion volume.

Overall, this work contributes to computational social science by moving beyond aggregate sentiment trends to show how emotions and topics are shaped by community norms, interaction structures, and practical use cases. By analyzing both posts and comments across multiple subreddits, the study demonstrates that public perception of generative AI is not uniform but emerges through collective negotiation within online communities. These findings underscore the importance of community-level analysis for understanding how emerging technologies are adopted, evaluated, and normalized. As generative AI continues to evolve, studying online discourse will remain critical for anticipating public concerns, guiding responsible AI development, and understanding how technological change is experienced in everyday digital life.

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